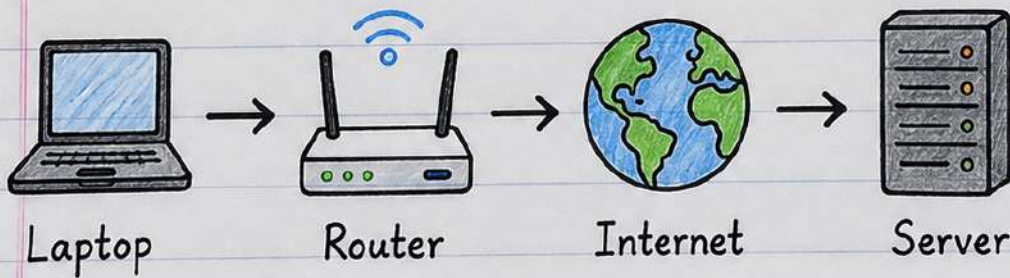


IP Address

≡ Explained Simply ≡

★ Every device on a network
needs an address. ★





What is an IP Address?



- IP stands for Internet Protocol.
- A unique address assigned to a device.
- Helps identify a device on a network.
- Used to send and receive data.



Think:
IP Address =
Home Address
of a Device



Laptop



Router



Internet



Server

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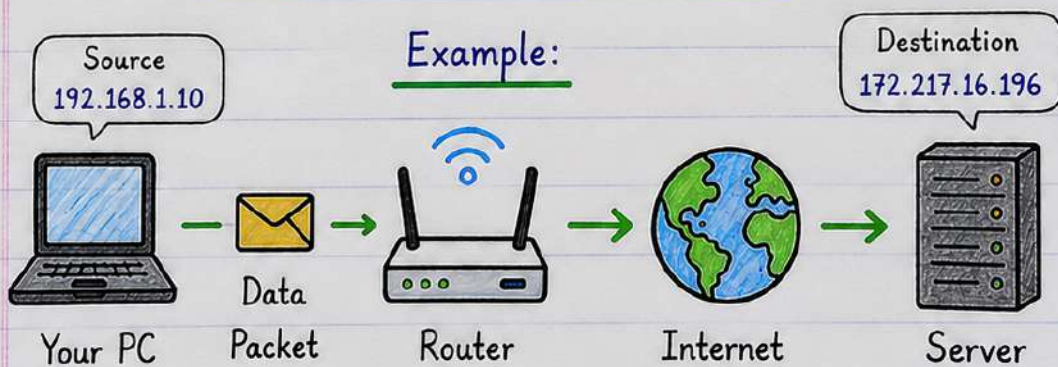
Why Do We Need IP Address?



- Without an IP address, data will not know where to go.

No IP =
No Delivery!

IP tells the network
WHERE to deliver data.



→ IP Address helps data reach the right device on the right network.



Types of IP Addresses



Two main versions:

IPv4

- Uses 32 bits
- Written in decimal format
- Example:

192.168.1.10

IPv6

- Uses 128 bits
- Written in hexadecimal format
- Example:

2001:0db8:85a3::8a2e:0370:7334



Think:

IPv4 and IPv6 do the same job, different formats!



Device



Router



Internet





IPv4 Explained



- IPv4 uses 32 bits.
- An IPv4 address is written in decimal format.
- It is divided into 4 parts separated by dots.



Key Point:

IPv4 has 4 sections.



Example:

192.168.1.10



192

168

1

10

Each part can have a value from 0 to 255.



Total: 4 parts $\times$ 8 bits = 32 bits
→ That's why IPv4 is 32-bit.





Public vs Private



IP Address



Private IP

- Used inside a local network (LAN).
- Not reachable directly from the Internet.
- Assigned by Router using DHCP.

Example:

192.168.1.10



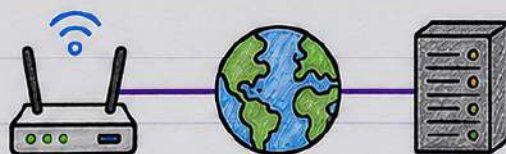
Inside Your Network

Public IP

- Used to communicate over the Internet.
- Reachable from anywhere on the Internet.
- Provided by ISP.

Example:

203.0.113.25



Across the Internet



Private IPs are like addresses inside your building,
Public IP is like your building's address on the road.



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Static vs Dynamic IP Address



Static IP

- Doesn't normally change.
- Manually configured
- Useful for servers.
- More stable and secure.
- Example: Web server, Mail server, etc.



Dynamic IP

- Changes automatically.
- Assigned by DHCP server
- Can change over time
- Used in most home / office networks.
- Less stable but easier to manage.



Think:

Static IP = Fixed address

Dynamic IP = Always moving address



Static IP



192.168.1.10

192.168.1.1

IP remains the same.

Dynamic IP



192.168.1.15

192.168.1.25

IP can change anytime.



Classes of IP Addresses



- IPv4 addresses are divided into 5 classes based on the first octet (first number).

Class	First Octet Range	Default Subnet Mask	Use
A	1 - 126	255.0.0.0	Large Networks
B	128 - 191	255.255.0.0	Medium Networks
C	192 - 223	255.255.255.0	Small Networks
D	224 - 239	N/A (Multicast)	Multicast Groups
E	240 - 255	N/A (Reserved)	Experimental Use



Remember:

- A, B, C → Unicast
- D → Multicast
- E → Reserved



Example:

- Class A → 10.0.0.1
- Class B → 172.16.0.1
- Class C → 192.168.1.1



Different classes, different sizes, same purpose - connect the world! ♥

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Private IP Ranges

(Not routed on Internet)



Class	Private IP Range	Used For
A	10.0.0.0 – 10.255.255.255	Large Organizations
B	172.16.0.0 – 172.31.255.255	Medium Organizations
C	192.168.0.0 – 192.168.255.255	Home / Small Networks



Key Points:

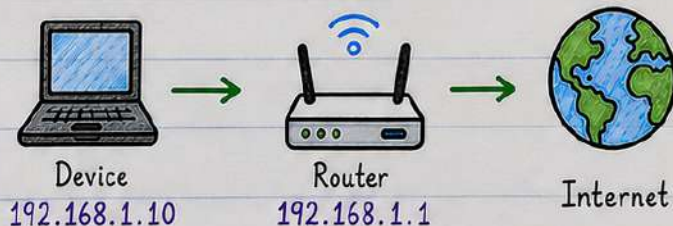
- Private IPs are used inside local networks.
- They cannot be reached directly from the Internet.
- NAT (Network Address Translation) is used to access Internet.



Think:

Your home Wi-Fi uses Private IPs.
Your router gives you Internet access.

Example (Home Network)



Note:

Private IP works inside network.
Public IP works outside (Internet).



Understanding IP addresses is the first step in Networking!

★ Quick Summary



- IP Address is a **unique** identifier for devices on a network.
- It helps devices **find** and communicate with each other.
- IPv4 uses **32-bit** addresses.
- IP Addresses can be **Public** or **Private**.
- Private IPs are used inside **local** networks.
- Public IPs are used to connect to the **Internet**.



Key Takeaway

IP Address is like our address in the digital world — it helps devices connect, communicate and stay connected.



Remember:

Same purpose, different classes and different uses! 😊

The Journey of Data



Your Device



Router



Internet



Destination Server



Understand today, Connect tomorrow! ★